NATIONAL ECONOMICS UNIVERSITY

FACULTY OF ARTIFICIAL INTELLIGENCE

KHOA TRÍ TUỆ NHÂN TẠO

MID-TERM ASSIGNMENT

Employee Churn Classification

Mid-term Project Classification:
Employee Churn Prediction

Group 7

Vũ Ngọc Minh Đức	MSV: 11252050
Vũ Quang Huy	MSV: 11253614
Nguyễn Ngọc Trường Sơn	MSV: 11257405

Course: Data Analysis with Python

Instructor: Trần Đức Minh

Abstract

This project predicts recorded employee churn using an HR dataset with $14{,}999$ rows and nine input features. After removing $3{,}008$ duplicate rows, we split the remaining $11{,}991$ records into training, validation, and test sets. Exploratory analysis shows that churn is linked to satisfaction, workload, and tenure in non-linear ways. For example, churn is high for both very few and very many projects. We compare Logistic Regression, KNN, Random Forest, and Gradient Boosting using five-fold cross-validation. We select by Left-class F1, because a model that always predicts *Stayed* gets 83% accuracy but finds no leavers. Random Forest is selected and reaches a test F1 of 0.95 (precision 0.98, recall 0.92), close to Gradient Boosting and far above Logistic Regression (0.55). The test set was seen before and the data have no timestamps, so the results show patterns in this dataset, not future churn.

Contents

Abstract i

1 Introduction and Problem Definition 1

1.1 Background and Motivation . . . 1
1.2 Problem Definition . . . 1
1.3 Objectives . . . 1

2 Data Description and Preparation 1

2.1 Data Source . . . 1
2.2 Data Inspection and Loading . . . 1
2.3 Data Dictionary . . . 2
2.4 Missing Values and Data Types . . . 2
2.5 Duplicate Removal . . . 2
2.6 Stratified Train/Validation/Test Split . . . 2

3 Exploratory Data Analysis 3

3.1 Dataset Overview and Target Distribution . . . 3
3.2 Univariate Analysis and Outliers . . . 4
3.3 Satisfaction and Evaluation . . . 4
3.4 Workload . . . 5
3.5 Tenure and Promotion . . . 5
3.6 Salary, Department and Work Accidents . . . 6
3.7 Correlation Analysis . . . 7
3.8 Key Insights . . . 7

4 Feature Selection and Preprocessing 8

4.1 Feature Roles . . . 8
4.2 Evidence for Feature Selection . . . 8
4.3 Preprocessing Pipeline . . . 8

5 Models and Experimental Setup 8

5.1 Classification Models . . . 8
5.2 Evaluation Metrics and Selection Criterion . . . 8
5.3 Overall Workflow . . . 10
5.4 Hyperparameter Tuning . . . 10

6 Results and Discussion **10**

6.1 Baseline . . . 10

6.2 Cross-validation Results (Training Set) . . . 11

6.3 Threshold Selection (Validation Set) . . . 11

6.4 Test-set Results (Previously Inspected Holdout) . . . 12

6.5 Confusion Matrix and Error Analysis . . . 12

6.6 Development and Test Comparison . . . 13

6.7 Feature Importance . . . 13

7 Conclusion **14**

7.1 Summary of Findings . . . 14

7.2 Practical Implications . . . 15

7.3 Limitations . . . 15

7.4 Future Improvements . . . 15

Data and Project Resources **15**

A Repository Structure and Reproducibility **16**

B Member Contributions **16**

C Supplementary Experimental Details **17**

C.1 Hyperparameter Search Grids . . . 17

1 Introduction and Problem Definition

1.1 Background and Motivation

Employee churn (turnover) is the loss of employees from an organization. This project examines which recorded employee attributes distinguish employees who left from those who stayed, and compares classification models on the group's HR dataset. The aim is to identify useful predictive patterns and assess how reliably the models detect the Left class.

1.2 Problem Definition

Task: binary classification of the recorded churn status of an employee.

Target variable: left $\in \{0,1\}$ (0 = Stayed, 1 = Left).

Input features: nine attributes of the employee (satisfaction, last evaluation, number of projects, monthly hours, tenure, work accident, promotion, department, salary level).

Scope note: the task classifies churn that has already been recorded. Predicting future departures would additionally require timestamps, a prediction horizon, and verification that each feature is known before an employee leaves.

1.3 Objectives

1. Inspect, clean, and split the data in a reproducible and leakage-free way.
2. Explore the data to identify features that are related to churn.
3. Build several classifiers, compare them with a baseline under one consistent protocol, and select a model and decision threshold without using the test set.
4. Evaluate and interpret the selected model using baseline comparison, a confusion matrix, and ROC and precision-recall curves.
5. Discuss practical implications, limitations and future work.

2 Data Description and Preparation

2.1 Data Source

The analysis uses one table, data/HR_Dataset.csv, with 14,999 rows and 10 columns. The exact dataset used by the group is available in the project repository (HR_Dataset.csv).

2.2 Data Inspection and Loading

The analysis uses the single HR_Dataset.csv table linked above, with nine input attributes and the target left. No table merging or concatenation is required. The file has no employee identifier, which limits verification of duplicate records.

2.3 Data Dictionary

Column	Type	Description
satisfaction_level	numeric (0.09–1.00)	Employee satisfaction level
last_evaluation	numeric (0.36–1.00)	Score of the last performance evaluation
number_project	integer (2–7)	Number of projects assigned
average_montly_hours	integer (96–310)	Average monthly working hours (montly is the original spelling)
time_spend_company	integer (2–10)	Years at the company
Work_accident	binary	1 = recorded work accident
promotion_last_5years	binary	1 = promoted in the last five years
Departments	categorical (10)	IT, RandD, accounting, hr, management, marketing, product_mng, sales, support, technical
salary	ordinal (3)	low, medium, high
left	binary	Target: 0 = Stayed, 1 = Left

Table 1: Data dictionary with observed ranges in the full dataset.

2.4 Missing Values and Data Types

The full dataset has no missing values, so no imputation is needed. Satisfaction and evaluation are stored as float64; projects, hours, tenure, binary flags, and the target are int64. Departments and salary are strings, encoded in the preprocessing pipeline (Section 4.3). There are no date columns.

2.5 Duplicate Removal

The group removed 3,008 exact duplicate rows in split_data.py, reducing the dataset from 14,999 to 11,991 records. Deduplication precedes splitting to prevent identical rows from appearing in both training and evaluation sets.

Of the removed rows, 1,580 (52.5%) belong to the Left class. The churn rate therefore falls from 23.8% (3,571 of 14,999) to 16.6% (1,991 of 11,991). Without employee identifiers, identical rows cannot be proved to represent the same person; all reported scores refer to this deduplicated dataset. No remaining rows have identical features but different labels.

2.6 Stratified Train/Validation/Test Split

Two calls to train_test_split, stratified on left with random_state=42, allocate 70% to training and divide the remaining 30% equally between validation and test. This maintains approximately the same minority-class proportion in all three sets.

Split	Rows	Left	Churn rate	Purpose
Training	8,393	1,394	16.61%	EDA, cross-validation, model fitting
Validation	1,799	299	16.62%	Threshold selection, permutation importance
Test	1,799	298	16.56%	Reporting model and baseline performance

Table 2: Fixed stratified splits (seed 42), containing all 11,991 deduplicated rows without overlap.

Training data support model fitting and cross-validation. After model selection, validation data determine the threshold and support permutation importance. Test scores are reported separately and do not determine the chosen model or threshold.

3 Exploratory Data Analysis

EDA examines distributions, outliers, and relationships with churn to inform feature and model choices. All feature-level analysis uses the training split only; only the dataset overview uses the full data.

Note. Feature-level patterns in this section are simple associations from the training set; the dataset overview uses the full data. None is read as a cause.

3.1 Dataset Overview and Target Distribution

Dataset	Employees	Left	Churn rate
Before duplicate removal	14,999	3,571	23.8%
After duplicate removal	11,991	1,991	16.6%

Table 3: Effect of duplicate removal on the target.

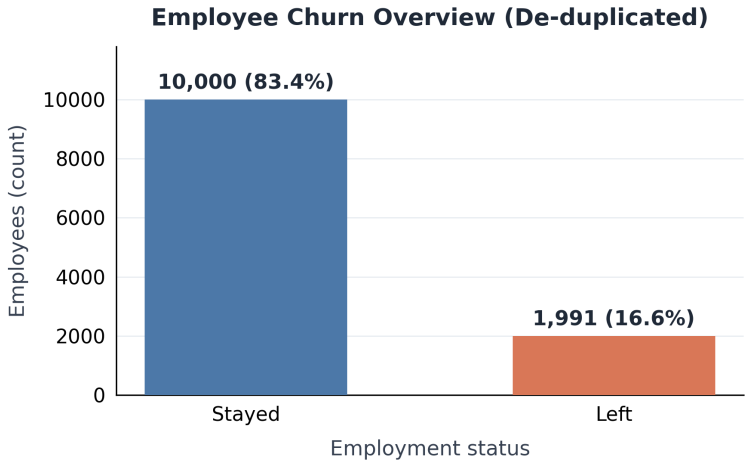


Figure 1: Distribution of the target after duplicate removal.

Interpretation. Stayers outnumber leavers about 5:1 (83.4% vs. 16.6%). A model that always predicts *Stayed* gets about 83% accuracy but finds no leavers, so we select models by Left-class F1 and use stratified splits.

3.2 Univariate Analysis and Outliers

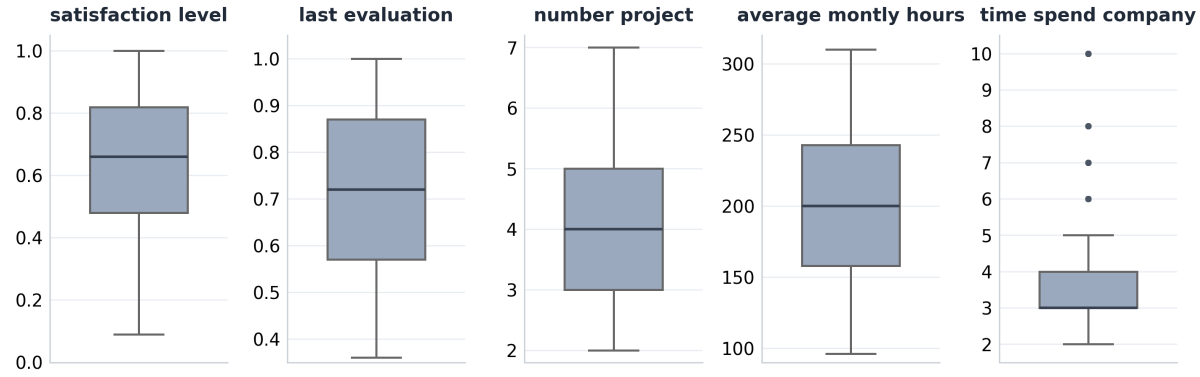


Figure 2: Boxplots of the numeric features (training set).

Interpretation. Only tenure has outliers under the $1.5 \times$ IQR rule: 590 rows (7.0%) at 6, 7, 8, or 10 years. These are recorded, plausible values, so we keep them; an outlier flag alone does not establish a data error. No training records with 7+ years are labeled Left (Section 3.5).

3.3 Satisfaction and Evaluation

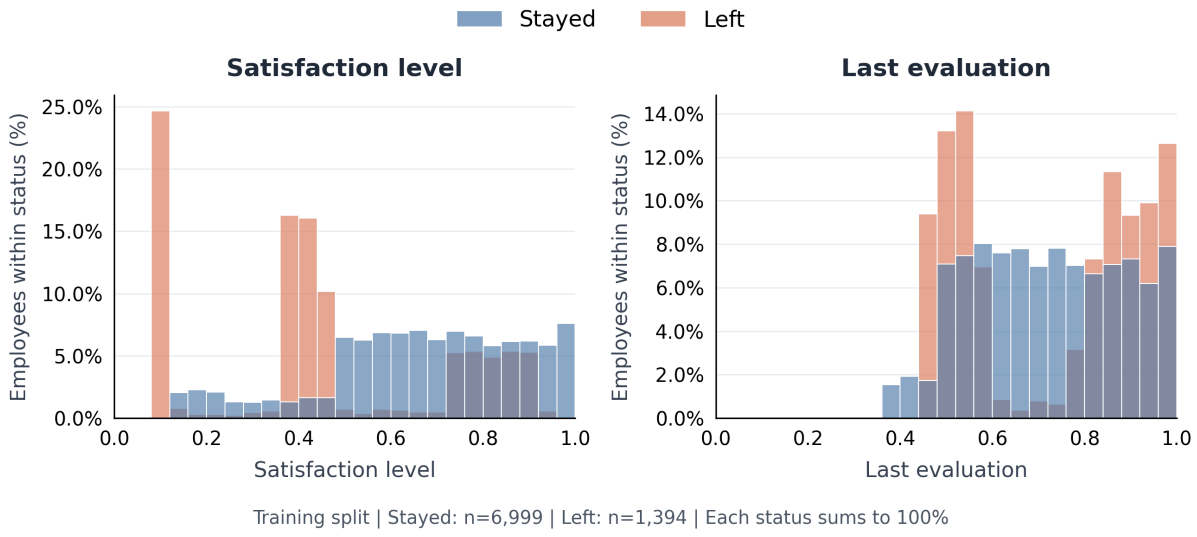


Figure 3: Satisfaction and evaluation by status; each status sums to 100%.

Interpretation. Leavers are not one group. Their satisfaction has three clusters ($\approx 26\%$ below 0.2, $\approx 44\%$ in $[0.3, 0.5)$, $\approx 27\%$ at least 0.7), so low satisfaction does not characterize all churn. Evaluation has almost the same mean for both groups (0.723 vs. 0.716), but leavers are split into low scores (43.7% below 0.6) and high scores (50.5% at least 0.8). Only 5.8% are in the middle $[0.6, 0.8)$ (stayers: 37.2%). Means and linear correlation can miss these patterns, motivating non-linear models.

3.4 Workload

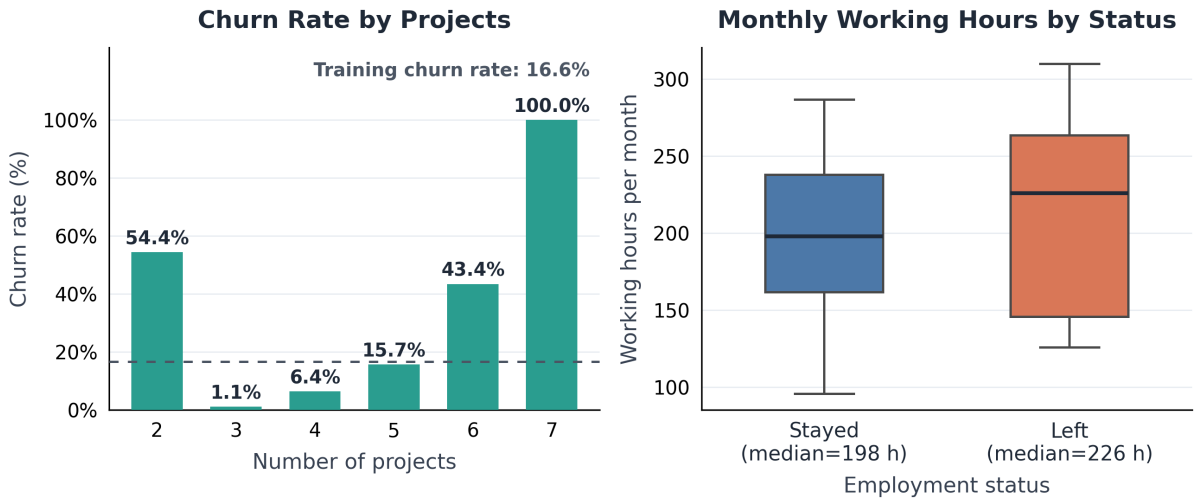


Figure 4: Churn rate by number of projects (left) and monthly working hours by status (right).

Interpretation. Both low and high workload are associated with more churn. Churn is 54.4% at 2 projects, 1.1–6.4% at 3–4, 43.4% at 6 and 100% at 7 ($n = 100$). Hours agree: 36.7% of leavers work at least $250\,\text{h}$ (stayers: 17.8%) and 30.3% at most $150\,\text{h}$ (stayers: 16.8%). The implemented Logistic Regression has no non-linear terms to directly fit this U-shape, which may partly explain its weaker result.

3.5 Tenure and Promotion

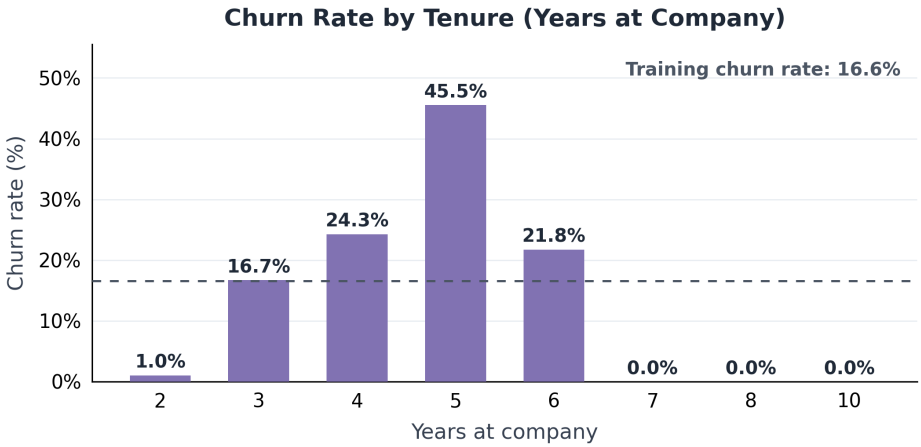


Figure 5: Churn rate by years at the company (training set).

Interpretation. Churn is concentrated in years 3–6 and peaks at 45.5% in year 5; year 2 also has 1.0% churn. No training records with $7+$ years are labeled Left ($n = 190$). Promoted employees have lower churn (5.1% vs. 16.8%), but only 7 of 136 left, so this small-group association should be interpreted cautiously.

3.6 Salary, Department and Work Accidents

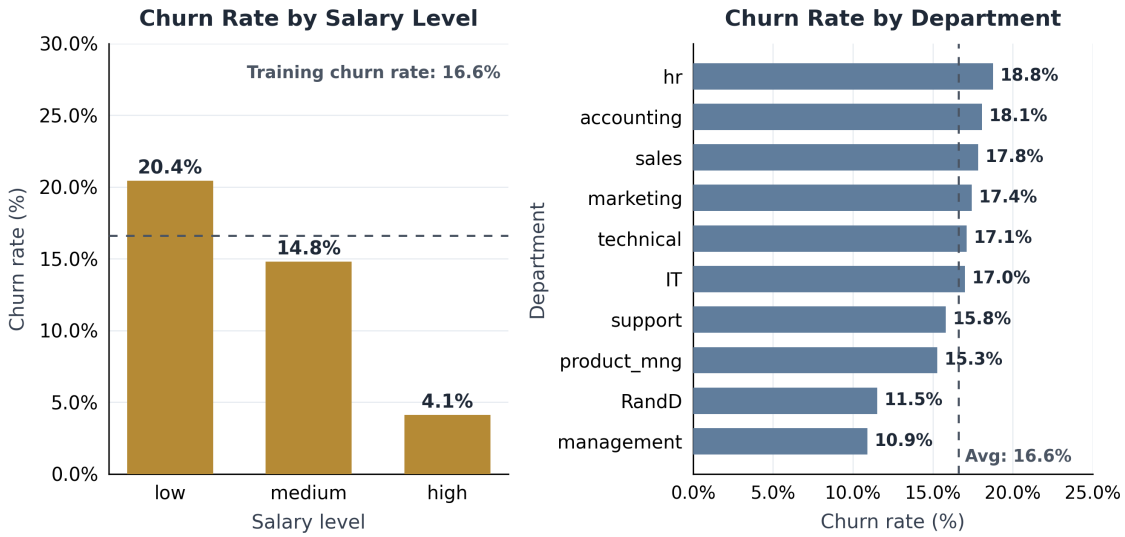


Figure 6: Churn rate by salary level (left) and by department (right); the dashed line is the training average of 16.6%.

Interpretation. Churn falls as salary rises (20.4%, 14.8%, 4.1%). Departments differ little (mostly 15–19%). Recorded work accidents are associated with lower churn (5.7% vs. 18.6%), but this does not show that an accident protects against leaving. This is another reason not to read these rates as causes.

3.7 Correlation Analysis

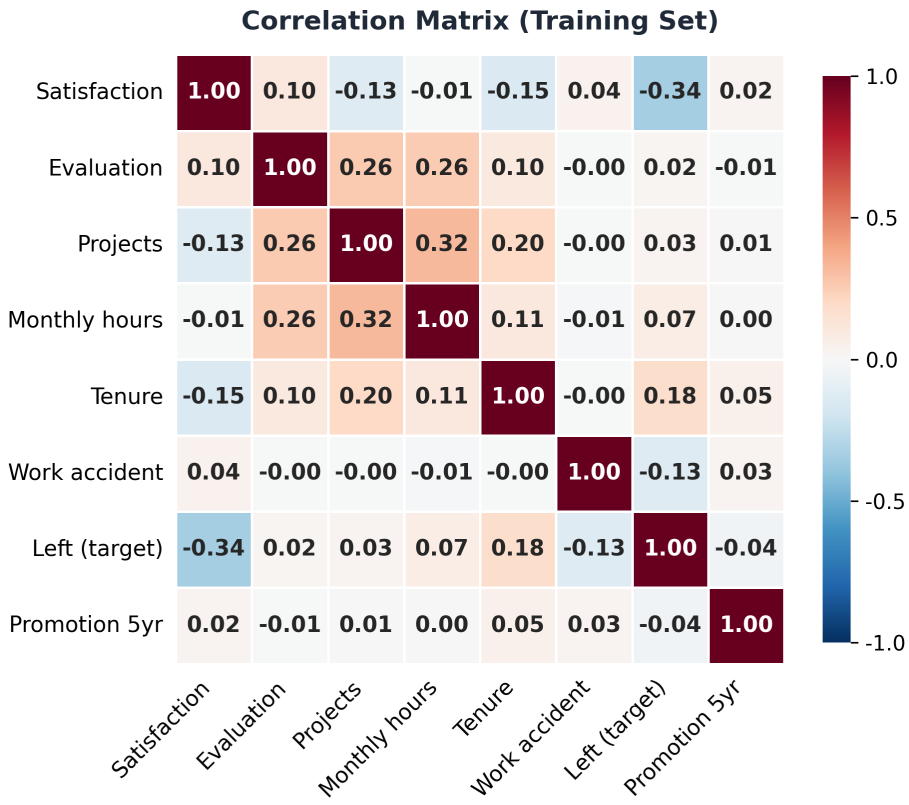


Figure 7: Pearson correlation matrix of numeric and binary columns (training set).

Interpretation. Linear correlations with `left` are limited: -0.343 for satisfaction, $+0.179$ for tenure, -0.126 for work accident, and $|r| < 0.08$ for the other numeric/binary inputs. Yet evaluation, projects and hours show non-linear patterns, so a linear-correlation filter could discard useful inputs. This supports retaining all nine features rather than filtering by correlation alone. Pairwise correlations among the numeric/binary inputs are modest (largest 0.324); this does not assess the encoded categorical inputs.

3.8 Key Insights

- Satisfaction, workload extremes, and mid-tenure distinguish leavers from stayers.
- Evaluation and workload show non-linear patterns that simple averages or linear correlations can obscure.
- Some subgroups have nearly deterministic outcomes, helping explain high classification scores within this dataset.

4 Feature Selection and Preprocessing

4.1 Feature Roles

The target is left. The inputs are the nine other columns: five numeric, two binary, one categorical, and one ordinal.

4.2 Evidence for Feature Selection

Training-set EDA highlights satisfaction, number of projects, monthly hours, evaluation, and tenure as potentially useful inputs. The implemented workflow retains all nine features; it does not perform a separate feature-subset search. Validation permutation importance explains the selected model (Section 6.7) and is not used to remove features or retune the model.

4.3 Preprocessing Pipeline

Each candidate model has its own ColumnTransformer plus classifier Pipeline. The preprocessing is fitted inside each cross-validation fold, so nothing from the validation folds leaks into preprocessing.

Feature type	Treatment	Applied to
Salary (ordinal)	Ordinal encoding (low < medium < high; unseen = -1)	all models
Departments	One-hot encoding (unknown categories ignored)	all models
Numeric features	Standard scaling	Logistic Regression, KNN only
Binary features	Kept as 0/1	all models

Table 4: Preprocessing by feature type.

5 Models and Experimental Setup

5.1 Classification Models

Logistic Regression provides the linear benchmark, KNN predicts from neighboring training examples, Random Forest ensembles randomized bagged trees, and Gradient Boosting builds boosted trees sequentially. The U-shaped workload pattern and bimodal evaluation scores in Sections 3.3 and 3.4 motivate comparing these non-linear models with the linear benchmark.

5.2 Evaluation Metrics and Selection Criterion

Left is the positive class (1), and Stayed is the negative class (0). True positives (TP) and true negatives (TN) are correctly classified leavers and stayers; false positives (FP) are stayers predicted to leave, and false negatives (FN) are leavers predicted to stay.

Accuracy

The fraction of all records classified correctly. It can be misleading when classes are imbalanced.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Precision

Among employees predicted to leave, the fraction who actually left.

$$P = \text{Precision} = \frac{TP}{TP + FP}.$$

Recall

The fraction of actual leavers successfully detected.

$$R = \text{Recall} = \frac{TP}{TP + FN}.$$

F1 score

The harmonic mean of precision and recall, balancing false positives and false negatives.

$$F_1 = \frac{2PR}{P + R} = \frac{2TP}{2TP + FP + FN}.$$

ROC-AUC

The area under the receiver operating characteristic curve, which plots the true positive rate (TPR) against the false positive rate (FPR) as the threshold varies. It measures score-ranking performance; higher is better, and random ranking has an expected value of 0.5.

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(u)\, du, \qquad u = \text{FPR}, \quad \text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

Average precision (AP)

Precision averaged over increases in recall across score thresholds. It summarizes positive-class ranking performance and is not trapezoidal precision–recall area.

$$\text{AP} = \sum_{k=1}^{m} (R_k - R_{k-1}) P_k.$$

Here, P_k and R_k are precision and recall at successive score thresholds ordered by increasing recall, with $R_0 = 0$ and $R_m = 1$.

Accuracy, precision, recall, and F1 use thresholded predictions; ROC-AUC and AP use the predicted scores across thresholds. The implementation sets undefined precision or F1 to zero. Models and the final threshold are selected using Left-class F1, not test scores. The always-Stayed baseline demonstrates why accuracy alone is insufficient: accuracy 0.8344 but F1 zero (Section 6.1).

5.3 Overall Workflow

All main models share five stratified cross-validation folds on the training set. GridSearchCV tunes hyperparameters using Left-class F1, with preprocessing fitted separately inside each fold. The predicted class is $\hat{y} = \mathbf{1}[\hat{p} \geq t]$, where the threshold t is chosen on validation data after selecting the model.

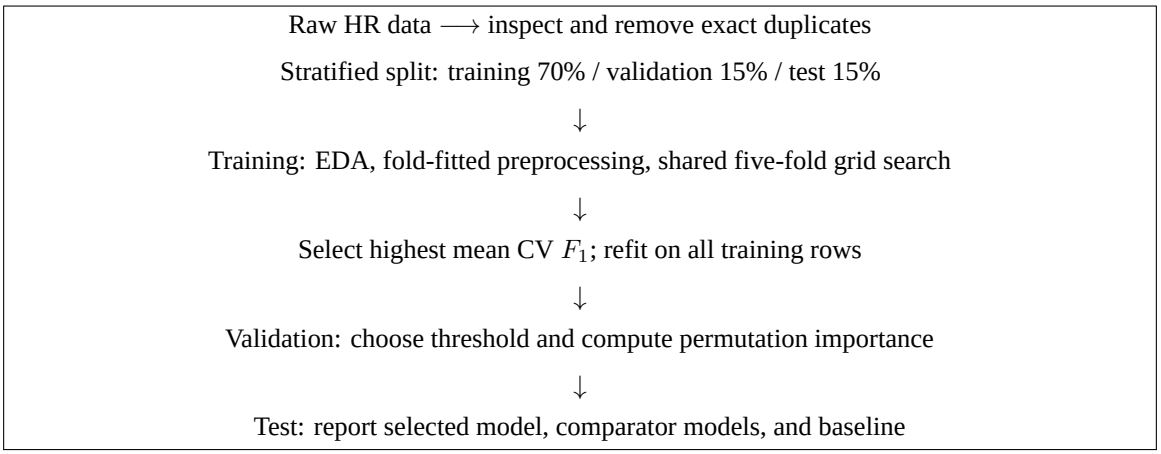


Figure 8: Overview of the modeling workflow.

1. Compare Logistic Regression, KNN, Random Forest, and Gradient Boosting with the same stratified training folds; select the highest mean CV F1 for Left.
2. Hyperparameters are tuned by GridSearchCV; model random seeds are fixed, not optimized.
3. Use the best pipeline already refitted on all training rows by GridSearchCV (its default refit behavior).
4. Choose the probability threshold on validation data: start at 0.5, try unique validation probabilities in ascending order, and replace the current threshold only for a strict F1 improvement. Ties retain the current threshold. Do not refit afterward.
5. Explain the selected model with validation permutation importance.
6. Report all tuned models on the test set at threshold 0.5, plus an always-Stayed baseline; report the selected model separately at its validation threshold. Test scores never choose the model or threshold.

5.4 Hyperparameter Tuning

GridSearchCV selects each model's settings using the shared training folds. The complete grids and best settings are in Appendix C.1; the selected Random Forest configuration is reported with its results.

6 Results and Discussion

6.1 Baseline

The always-Stayed baseline achieves accuracy 0.8344 (the test-set share of stayers), but recall and F1 are zero, precision is set to zero by the evaluation convention, and ROC-AUC is 0.5. It detects no departures.

Logistic Regression provides a learned benchmark with test F1 0.5513. Accuracy near 83% alone therefore does not demonstrate useful detection of the Left class.

6.2 Cross-validation Results (Training Set)

Model	Mean CV F1	Fold std. dev.
Random Forest	0.9451	0.0112
Gradient Boosting	0.9436	0.0133
KNN	0.8712	0.0103
Logistic Regression	0.5456	0.0066

Table 5: Five-fold stratified CV on the training set (tuned models).

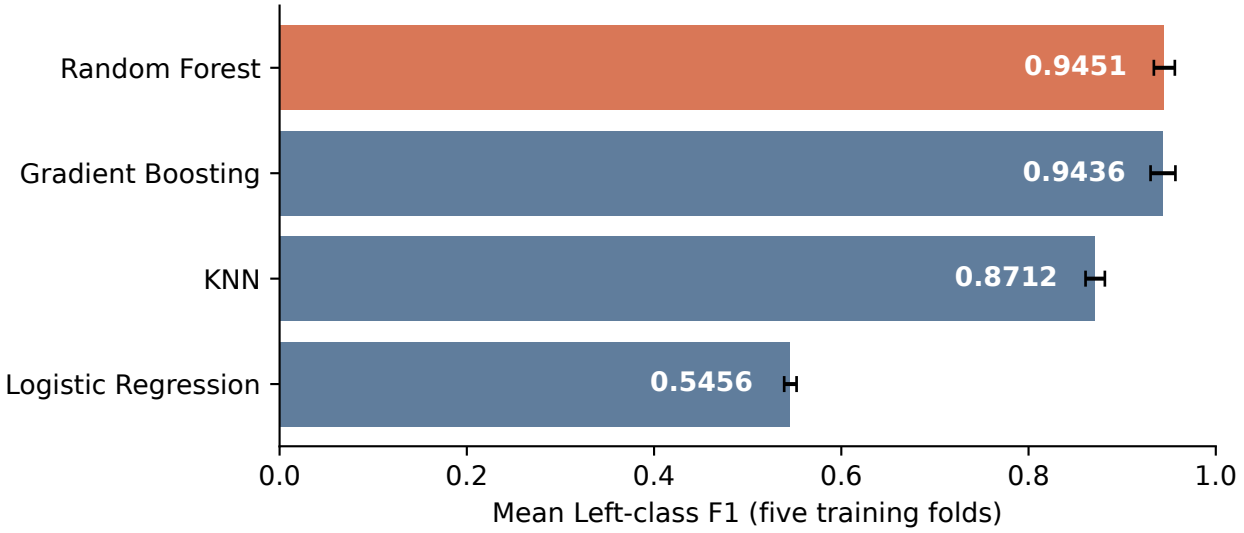


Figure 9: Mean cross-validation F1 of the Left class with fold standard deviations.

Interpretation. The tree ensembles beat KNN (0.871) and Logistic Regression (0.546) by a wide margin. Random Forest and Gradient Boosting differ by only 0.0015, less than the fold standard deviations (0.011–0.013). This is a small observed gap, but fold standard deviations alone do not establish equivalence or statistical significance. Random Forest is chosen because it ranks first under our preset rule.

6.3 Threshold Selection (Validation Set)

Starting from 0.5, we evaluated all unique validation probabilities and selected t = 0.4983 by validation F1. The change raises validation F1 from 0.9583 to 0.9601 (+0.0018), a small improvement over the default threshold.

Interpretation. The threshold $t = 0.4983$ lifts validation F1 from 0.9583 to 0.9601, a very small gain. The default threshold 0.5 already gives similar performance; these two scores alone do not establish that F1 is flat throughout the surrounding threshold range.

6.4 Test-set Results (Previously Inspected Holdout)

Model	Thr.	Acc.	Prec.	Recall	F1	ROC-AUC	AP
Always Stayed (baseline)	0.5	0.8344	0.0000	0.0000	0.0000	0.5000	0.1656
Logistic Regression	0.5	0.7765	0.4130	0.8289	0.5513	0.8453	0.3843
KNN	0.5	0.9566	0.8374	0.9161	0.8750	0.9532	0.8531
Random Forest	0.5	0.9828	0.9785	0.9161	0.9463	0.9748	0.9611
Gradient Boosting	0.5	0.9844	0.9821	0.9228	0.9516	0.9753	0.9527
Selected RF (val. threshold)	0.4983	0.9839	0.9786	0.9228	0.9499	0.9748	0.9611

Table 6: Test-set performance (Left is the positive class).

Interpretation. The tree ensembles are best (F1 ≈ 0.95), then KNN (0.875) and Logistic Regression (0.551). Logistic Regression's accuracy (0.777) is lower than the baseline's (0.834), so accuracy alone would rank it below a model that finds no leavers. Selected-threshold Random Forest and default-threshold Gradient Boosting differ by about 0.0017 F1 and disagree on one test case, so we keep the CV choice. Because the test set was seen before, these scores may be optimistic.

6.5 Confusion Matrix and Error Analysis

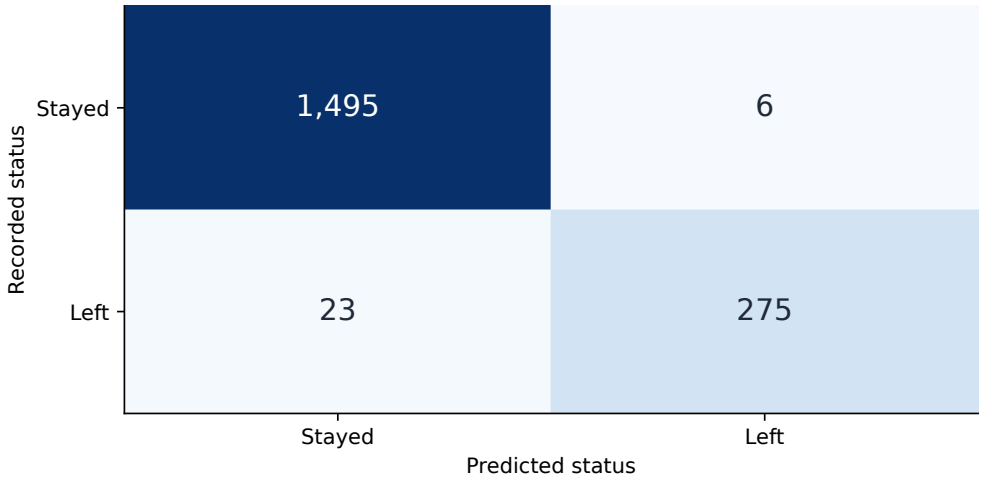


Figure 10: Confusion matrix of the selected Random Forest on the test set.

The model misses 23 of 298 leavers (7.7%) and wrongly flags 6 of $1{,}501$ stayers (0.4%). If a missed leaver costs more than an extra retention talk, a lower threshold should be tested on development data once costs are known.

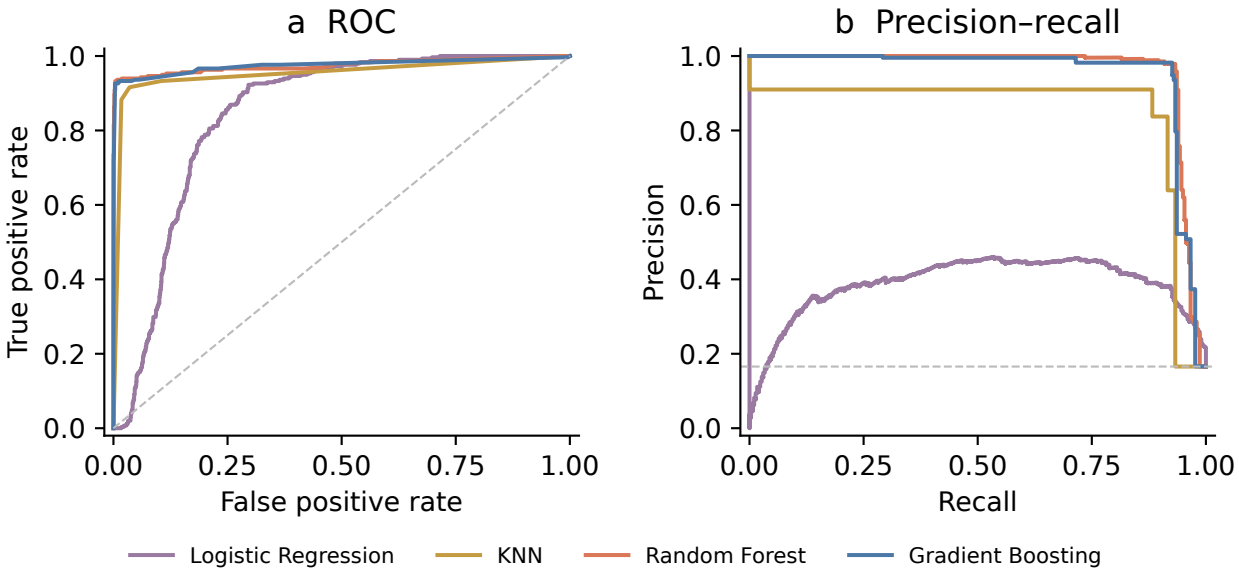


Figure 11: ROC (left) and precision-recall (right) curves of the four main models on the test set.

Interpretation. The Random Forest and Gradient Boosting curves are close over much of the range. Logistic Regression has ROC-AUC 0.845 but AP only 0.384 (the no-skill precision reference is the churn prevalence, 0.166). With $1{,}501$ stayers, even a low false-positive rate can produce appreciable false alarms. ROC does not directly show the fraction of flagged employees who are false alarms; precision does.

6.6 Development and Test Comparison

Random Forest has mean training-set CV F1 0.9451 at the default decision rule, validation F1 0.9601 at the selected threshold, and test F1 0.9499 at that threshold. These are not directly comparable estimates: CV also selects hyperparameters, validation selects the threshold, and test was inspected earlier. No resubstitution training F1 is reported by the implemented workflow.

6.7 Feature Importance

Permutation importance shuffles one original feature at a time on the validation set and measures the drop in churn-class F1 at the selected threshold (10 repeats).

Feature	Mean F1 decrease	Std.
satisfaction_level	0.6245	0.0186
last_evaluation	0.3598	0.0202
number_project	0.3516	0.0169
average_montly_hours	0.3471	0.0123
time_spend_company	0.2912	0.0173
Departments	0.0015	0.0015
Work_accident	0.0003	0.0007
promotion_last_5years	0.0002	0.0005
salary	0.0002	0.0005

Table 7: Validation permutation importance of the selected Random Forest.

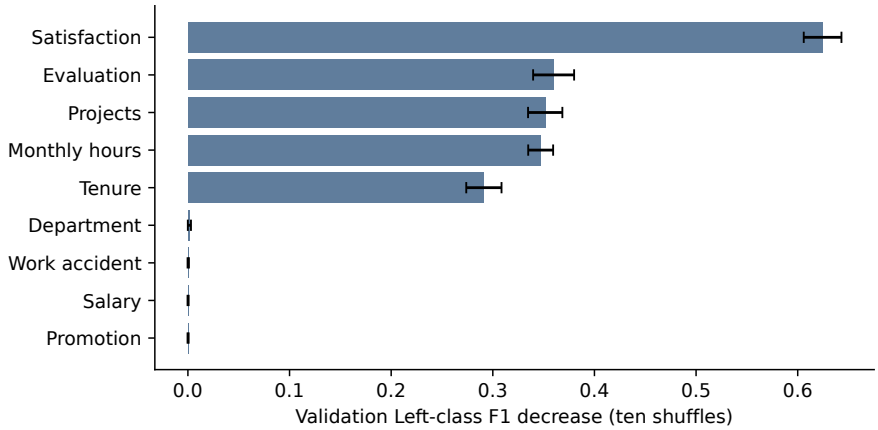


Figure 12: Permutation importance on the validation set (error bars: standard deviation over 10 repeats).

Interpretation. Satisfaction matters most to this model (F1 drop 0.62), then evaluation, projects and hours (≈ 0.35) and tenure (0.29), matching the EDA. These drops are not additive: their sum exceeding the model's F1 does not by itself prove feature interactions. Salary, department, accident and promotion are near zero, even though salary has a clear gradient in EDA ($20.4\% \rightarrow 4.1\%$). The main features may already carry some of this information, but this analysis does not establish that explanation. These are predictive links, not causes.

7 Conclusion

7.1 Summary of Findings

Churn in this dataset is linked to low or mixed satisfaction, very low or very high workload (2 vs. 6–7 projects; at most 150 or at least 250 hours), mid tenure (years 3–6), and low salary. Most linear correlations with churn are small, but several non-linear patterns are strong.

The best model is Random Forest (100 trees, `max_depth=None`, `min_samples_split=5`), selected by CV F1 (0.9451) and used at threshold 0.4983. On the test set it reaches accuracy 0.9839, precision 0.9786, recall

0.9228, and F1 0.9499. All learned models find leavers, while the always-Stayed baseline finds none. The two tree ensembles (Random Forest and Gradient Boosting, F1 ≈ 0.95) are clearly better than KNN (0.875) and Logistic Regression (0.551), and they differ from each other by only 0.0017 F1.

7.2 Practical Implications

The results show which employee features are most linked to churn in this dataset: satisfaction, workload, tenure, and salary. They also show that tree-based models detect leavers far better than a linear model on this data. The model is a predictive tool for this dataset only. It does not show why employees leave.

7.3 Limitations

- **No timestamps.** The model classifies recorded churn, not future churn. Satisfaction may have been measured after the decision to leave.
- **Test set reuse and size.** The test set was seen in an earlier version and has only 298 leavers, so recall is uncertain.
- **Deduplication.** Without employee IDs, removing duplicates changed the churn rate from 23.8% to 16.6%.
- **Easy subgroups.** Some groups are fully predictable (e.g. 100% churn at 7 projects), so scores may be higher than on other data.
- **Association only.** Feature importance shows prediction links, not causes.

7.4 Future Improvements

- Repeat the split with several random seeds, or use repeated cross-validation, to check that the model ranking and recall stay stable.
- Widen the search grids (some best values lie on the grid edge) and compare Random Forest and Gradient Boosting with a paired test on the same folds.
- Add a shallow Decision Tree and an XGBoost model, and check whether simple rules can reach similar scores.
- Try the same pipeline on the other HR dataset from the assignment to see whether the results hold on different data.

Data and Project Resources

- **Dataset used in the analysis:** HR_Dataset.csv, available from the group's repository: `https://github.com/ducvu0904/Employee-churn---Group-7/blob/main/data/HR_Dataset.csv`.
- **Group 7 project repository:** analysis notebooks, data preparation code, and saved results: `https://github.com/ducvu0904/Employee-churn---Group-7`.

A Repository Structure and Reproducibility

Public repository: https://github.com/ducvu0904/Employee-churn---Group-7.

The analysis is organized in a single notebook, Employee_Churn_Analysis.ipynb, using data/HR_Dataset.csv as its input. It covers data inspection, duplicate removal, stratified splitting, exploratory analysis, preprocessing, model comparison, and evaluation.

The repository includes requirements.txt for the software dependencies and README.md for usage details. Figures are stored in plots/, and the final test metrics are saved in results/test_metrics.csv. Supporting tables and interpretations appear within the notebook.

B Member Contributions

Member	ID	Contribution
Vũ Ngọc Minh Đức	11252050	Data cleaning, EDA visualization & descriptive analysis
Vũ Quang Huy	11253614	Machine learning pipeline, GridSearchCV tuning & threshold optimization
Nguyễn Ngọc Trường Sơn	11257405	Model evaluation, error analysis, report compilation & LaTeX formatting

Table 8: Member contributions, as listed in the source report.

C Supplementary Experimental Details

C.1 Hyperparameter Search Grids

All grids use GridSearchCV with Left-class F1, shared five-fold stratified cross-validation, and seed 42.

Model	Search grid	Best setting found
Logistic Regression	C ∈ {0.1, 1, 10}; class_weight ∈ {None, balanced}	C = 1; class_weight = balanced
KNN	k ∈ {3, 5, 7, 9}; weights ∈ {uniform, distance}	k = 3; weights = uniform
Random Forest	n_estimators ∈ {100, 200}; max_depth ∈ {10, 20, None}; min_samples_split ∈ {2, 5}	100 trees; max_depth = None; min_samples_split = 5
Gradient Boosting	n_estimators ∈ {100, 150}; learning_rate ∈ {0.001, 0.01, 0.1}; max_depth ∈ {3, 5}	100 trees; learning_rate = 0.01; max_depth = 5

Table 9: Full search grids and best settings.

Some best values lie on the edge of their search grid, including KNN k = 3 and Random Forest with 100 trees. A wider search is a possible future experiment, not an implemented result.